

Week 9

1. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear such that transformation

$$T \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \text{ and } T \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Find } T \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \text{ and } T \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$$

First write $u = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ as linear combinations of $\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 0 & 1 & 2 & 2 \\ -1 & 1 & 1 & 1 \\ 2 & -2 & -2 & -3 \end{array} \right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & -1 \end{array} \right]$$

$\rightarrow$ no solution $\rightarrow$ i.e. $\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ is not a linear combination of $\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$, hence $T \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ cannot be determined!

$$\text{But, } T \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = T \left(\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right) = T \left(\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \right) + 2T \left(\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$$

2. Find the matrix of the linear transformation obtained by a reflection in the Y-axis followed by rotation by π radians.

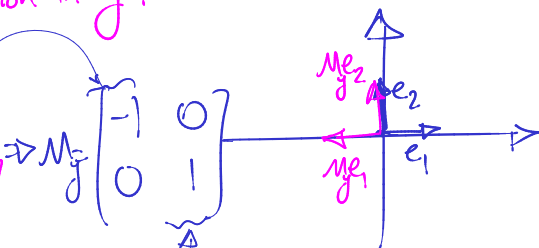
Reflection in y followed by rotation π radians:

$$R_{\pi} M_y = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

M_y : Reflection in y:

$$M_y e_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

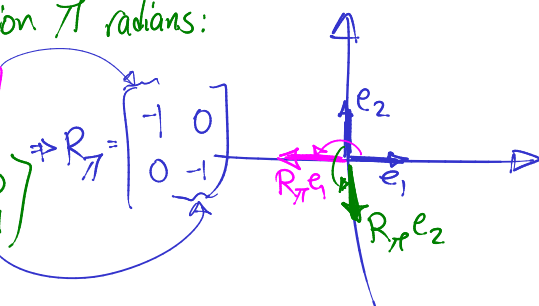
$$M_y e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



R_{π} : Rotation π radians:

$$R_{\pi} e_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$R_{\pi} e_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$



[Q: Does this look like a rotation or reflection? which one?]

3. Let T be a linear transformation induced by the matrix $A = \begin{bmatrix} 4 & -3 \\ 2 & -2 \end{bmatrix}$.

Find the matrix of T^{-1} .

$\hookrightarrow$ is $A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{4(-2) - (-3)2} \begin{bmatrix} -2 & 3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -3/2 \\ 1 & -2 \end{bmatrix}$

(check $A^{-1}A = I$: $A^{-1}A = \begin{bmatrix} 1 & -3/2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$)

(Question: Do we have to check $AA^{-1} = I$, too? why?)

4. Find the matrix for $T(\vec{w}) = \text{proj}_{\vec{v}}(\vec{w})$ where $\vec{v} = [1 \ -2 \ 3]^T$.

Let the matrix of T be P .

Sol 1: $P = \frac{V V^T}{V^T V} = \frac{\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} [1 \ -2 \ 3]}{1+4+9} = \frac{1}{14} \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ 3 & -6 & 9 \end{bmatrix}$.

Sol 2: $P = [T(e_1) | T(e_2) | T(e_3)]$

$T(e_1) = \text{Proj}_{\vec{v}}(e_1) = \frac{e_1 \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{1}{1+4+9} \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

$T(e_2) = \text{Proj}_{\vec{v}}(e_2) = \frac{e_2 \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{-2}{1+4+9} \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} -2 \\ 4 \\ -6 \end{bmatrix}$

$T(e_3) = \text{Proj}_{\vec{v}}(e_3) = \frac{e_3 \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{3}{1+4+9} \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 3 \\ -6 \\ 9 \end{bmatrix}$

$\Rightarrow P = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \\ 3 & -6 & 9 \end{bmatrix}$

5. Divide $\frac{3-i}{2-3i}$

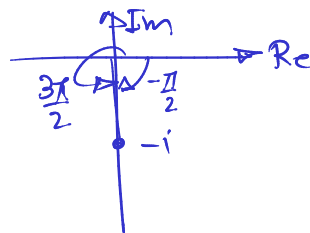
$$\frac{3-i}{2-3i} = \frac{(3-i)(2+3i)}{(2-3i)(2+3i)} = \frac{6+9i-2i-3i^2}{4+9} = \frac{9+7i}{13} = \frac{9}{13} + \frac{7}{13}i$$

6. Find the roots of $x^2 - 6x + 13 = 0$ and $x^2 + 2x + 1 + i = 0$

Roots of $x^2 - 6x + 13 = 0$ are $x = \frac{-(-6) \pm \sqrt{36 - 4 \cdot 1 \cdot 13}}{2} = \frac{6 \pm \sqrt{16}}{2} = 3 \pm 2i$.

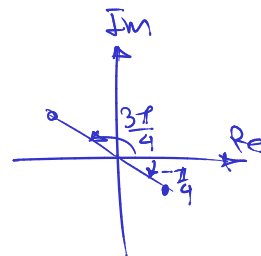
Roots of $x^2 + 2x + (1+i) = 0$ are $x = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot (1+i)}}{2} = \frac{-2 \pm 2\sqrt{-i}}{2} = -1 \pm \sqrt{-i}$ (*)

~> We have to find roots of $-i = 1e^{-\frac{\pi}{2}i} = 1e^{\frac{3\pi}{2}i}$
second



Let this root be $q = re^{i\theta} \Rightarrow q^2 = r^2 e^{2i\theta} = \begin{cases} 1e^{-\frac{\pi}{2}i} \\ 1e^{\frac{3\pi}{2}i} \end{cases}$

~> $\begin{cases} r=1 \\ 2\theta = -\frac{\pi}{2} \text{ or } \frac{3\pi}{2} \Rightarrow \theta = -\frac{\pi}{4} \text{ or } \frac{3\pi}{4} \end{cases} \Rightarrow \begin{cases} q = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i \\ \text{or} \\ q = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{cases} \Rightarrow q = \pm \frac{\sqrt{2}}{2}(1-i)$



(*) $\Rightarrow x = -1 \pm \frac{\sqrt{2}}{2}(1-i) \Rightarrow \begin{cases} x = \left(-1 + \frac{\sqrt{2}}{2}\right) - \frac{\sqrt{2}}{2}i \\ \text{or} \\ x = \left(-1 - \frac{\sqrt{2}}{2}\right) + \frac{\sqrt{2}}{2}i \end{cases}$